

Requirement Analysis

Data Flow Diagram & User Stories

Date	14-07-2026
Team ID	
Project Name	EduGenie
Maximum Marks	

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 (Context Level):

[Learner] — question / topic / text / subject → [EduGenie System] → answer / explanation / quiz / summary / roadmap → [Learner]

[EduGenie System] ↔ prompt / generated content ↔ [Generative AI Service (e.g., Gemini API)]

DFD Level 1 (Detailed):

Learner Input → Frontend (HTML/CSS/JS) → FastAPI Router, which then routes to one of five modules:

1. Router → Q&A Module → AI Service → Formatted Answer → Frontend
2. Router → Explanation Module → AI Service → Simplified Explanation → Frontend
3. Router → Quiz Generation Module → AI Service → Structured Quiz → Frontend
4. Router → Summarization Module → AI Service → Summary → Frontend
5. Router → Learning Path Module → AI Service → Structured Roadmap → Frontend

User Stories:

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Student	Question Answering	USN-1	As a student, I can ask a question and receive an instant, accurate answer with context	I receive a relevant answer within a few seconds	High	Sprint-1
Student	Concept Explanation	USN-2	As a student, I can request a simplified explanation of a concept I'm studying	I receive a plain-language explanation I can understand	High	Sprint-2
Student	Quiz Generation	USN-3	As a student, I can select "Generate Quiz" for a topic to self-evaluate	I receive topic-specific questions and can check my answers	High	Sprint-3
Self-learner	Summarization	USN-4	As a self-learner, I can paste long	I receive a shorter,	Medium	Sprint-3

			study material to get a concise summary	accurate summary of the original text	m	
Self-learner	Learning Path	USN-5	As a self-learner, I can request a personalized roadmap for a new subject	I receive a staged (beginner/intermediate/advanced) roadmap with recommendations	Medium	Sprint-4
Educator	Quiz Generation	USN-6	As an educator, I can generate practice questions on a topic for my students	I receive a usable set of quiz questions for the given topic	Medium	Sprint-3
Any User	Interactive UI	USN-7	As a user, I can access all five features from a single, responsive interface	I can navigate between Q&A, Explain, Quiz, Summarize, and Learning Path without confusion	High	Sprint-2